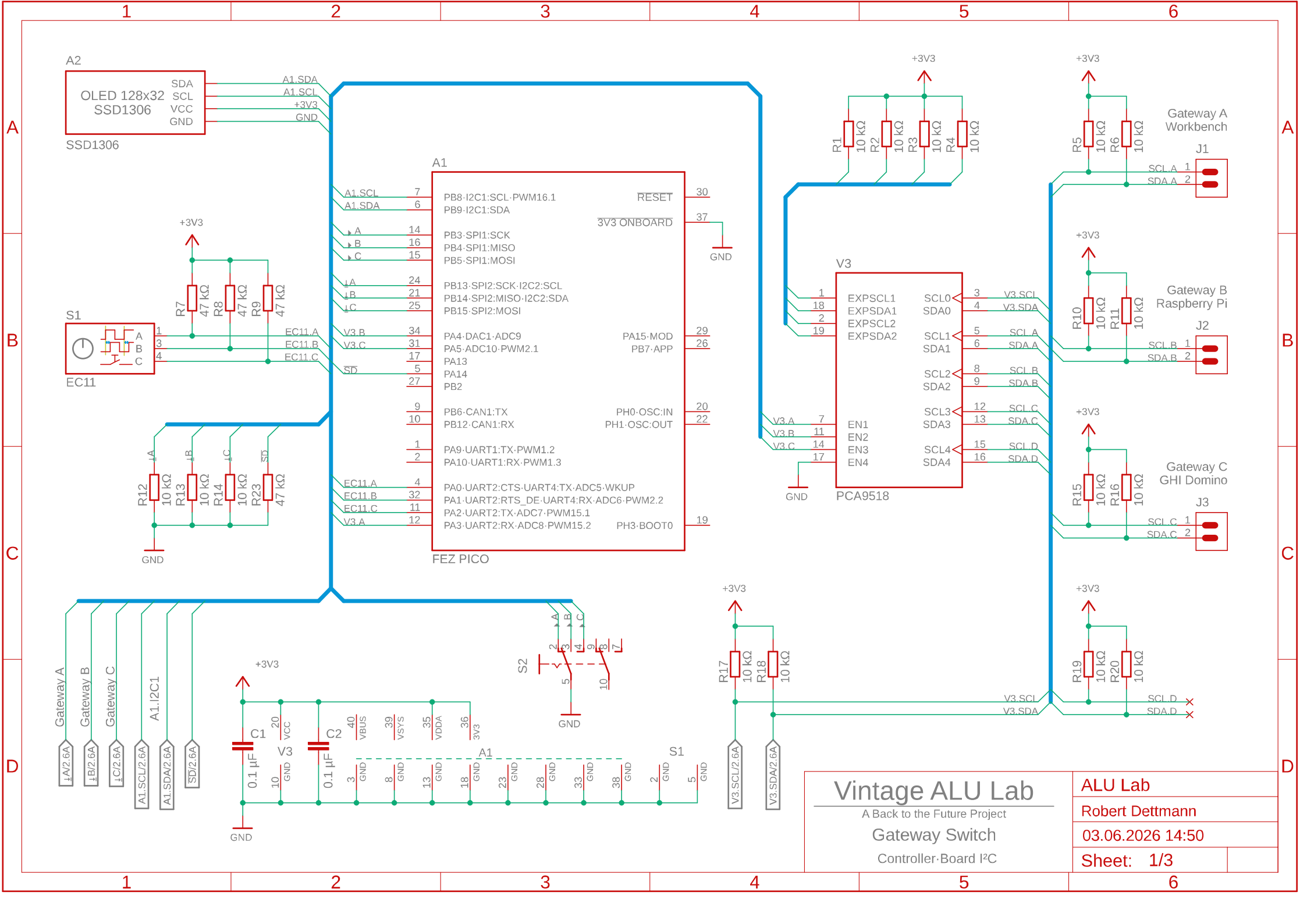
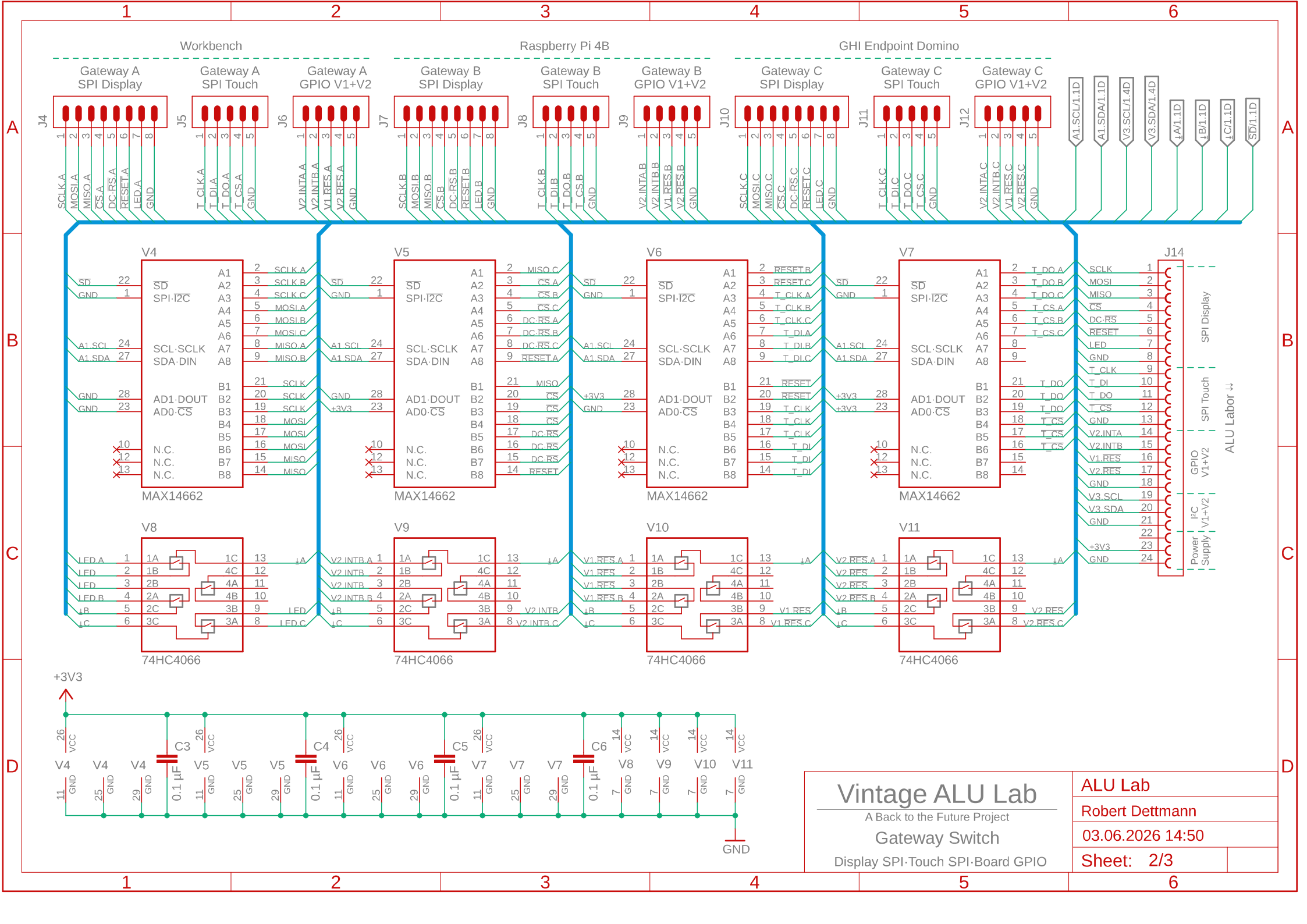

A2
OLED 128x32
SSD1306
SDA
SCL
VCC
GND
SSD1306
A1.SDA
A1.SCL
+3V3
GND
A1
PB8·I2C1:SCL·PWM16.1
PB9·I2C1:SDA
PB3·SPI1:SCK
PB4·SPI1:MISO
PB5·SPI1:MOSI
PB13·SPI2:SCK·I2C2:SCL
PB14·SPI2:MISO·I2C2:SDA
PB15·SPI2:MOSI
PA4·DAC1·ADC9
PA5·ADC10·PWM2.1
PA13
PA14
PB2
PB6·CAN1:TX
PB12·CAN1:RX
PA9·UART1:TX·PWM1.2
PA10·UART1:RX·PWM1.3
PA0·UART2:CTS·UART4:TX·ADC5·WKUP
PA1·UART2:RTS_DE·UART4:RX·ADC6·PWM2.2
PA2·UART2:TX·ADC7·PWM15.1
PA3·UART2:RX·ADC8·PWM15.2
RESET
3V3 ONBOARD
PA15·MOD
PB7·APP
PH0·OSC:IN
PH1·OSC:OUT
PH3·BOOT0
FEZ PICO
S1
EC11
EC11.A
EC11.B
EC11.C
R7 47 kΩ
R8 47 kΩ
R9 47 kΩ
R12 10 kΩ
R13 10 kΩ
R14 10 kΩ
R23 47 kΩ
R1 10 kΩ
R2 10 kΩ
R3 10 kΩ
R4 10 kΩ
V3
PCA9518
EXPSCL1
EXPSDA1
EXPSCL2
EXPSDA2
SCL0
SDA0
SCL1
SDA1
SCL2
SDA2
SCL3
SDA3
SCL4
SDA4
EN1
EN2
EN3
EN4
V3.SCL
V3.SDA
R5 10 kΩ
R6 10 kΩ
R10 10 kΩ
R11 10 kΩ
R15 10 kΩ
R16 10 kΩ
R17 10 kΩ
R18 10 kΩ
R19 10 kΩ
R20 10 kΩ
Gateway A
Workbench
Gateway B
Raspberry Pi
Gateway C
GHI Domino
J1
J2
J3
SCL_A
SDA_A
SCL_B
SDA_B
SCL_C
SDA_C
SCL_D
SDA_D
S2
C1 0.1 µF
C2 0.1 µF
Gateway A
Gateway B
Gateway C
A1.I2C1
A1.SCL/2.6A
A1.SDA/2.6A
Vintage ALU Lab
A Back to the Future Project
Gateway Switch
Controller·Board I²C
ALU Lab
Robert Dettmann
03.06.2026 14:50
Sheet: 1/3



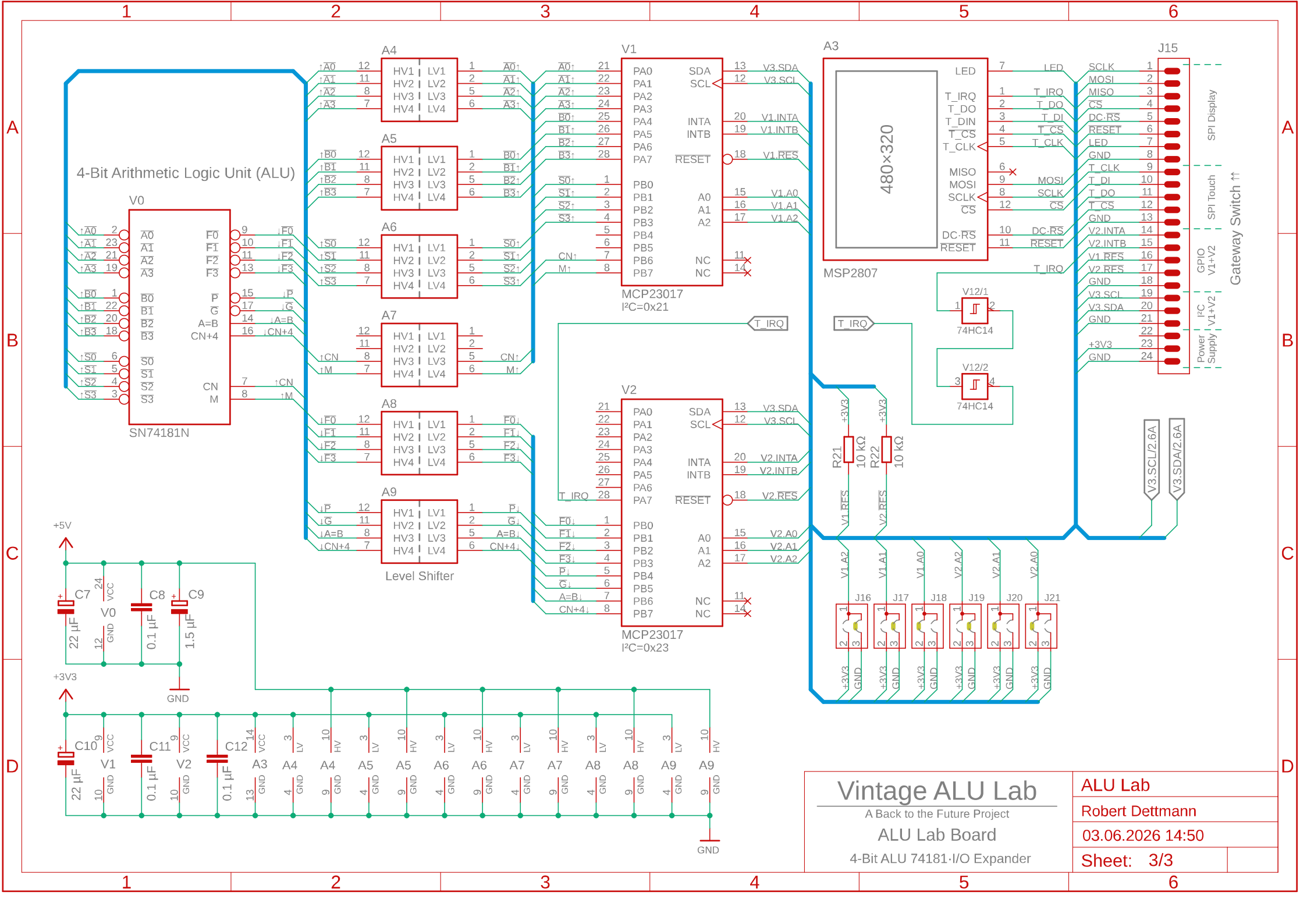
Vintage ALU Lab

A Back to the Future Project

Gateway Switch

Display SPI·Touch SPI·Board GPIO

ALU Lab	
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Sheet: 2/3	


4-Bit Arithmetic Logic Unit (ALU)
V0
SN74181N
A4
A5
A6
A7
A8
A9
HV1
HV2
HV3
HV4
LV1
LV2
LV3
LV4
Level Shifter
V1
MCP23017
I²C=0x21
V2
MCP23017
I²C=0x23
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
SDA
SCL
INTA
INTB
RESET
A0
A1
A2
NC
T_IRQ
A3
480×320
MSP2807
LED
T_IRQ
T_DO
T_DIN
T_CS
T_CLK
MISO
MOSI
SCLK
CS
DC·RS
RESET
J15
SPI Display
SPI Touch
GPIO V1+V2
I²C V1+V2
Power Supply
Gateway Switch ↑↑
V12/1
V12/2
74HC14
R21
R22
10 kΩ
+3V3
+5V
GND
C7
C8
C9
C10
C11
C12
22 µF
0.1 µF
1.5 µF
J16
J17
J18
J19
J20
J21
V3.SCL/2.6A
V3.SDA/2.6A
Vintage ALU Lab
A Back to the Future Project
ALU Lab Board
4-Bit ALU 74181·I/O Expander
ALU Lab
Robert Dettmann
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